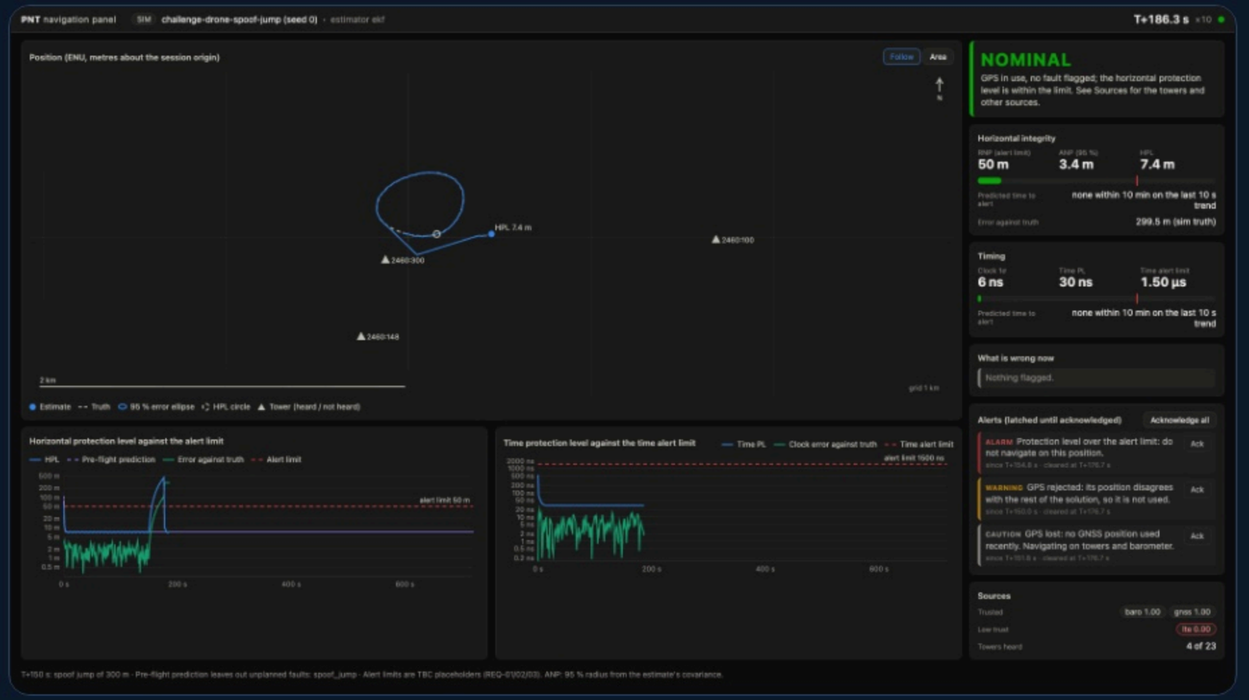
HACK THE HORIZONS · CHALLENGE 2

Green, confident, and 300 m wrong

The real danger is not losing GPS. It is a display that says “trust me” when it should not.

We built a navigation layer that knows when it is wrong.



Simulation, one run: a plain filter 27 s after a 300 m GPS jump

WHAT WE BUILT

A receive-only payload that records real cell towers, and a laptop that says when not to trust the position

PAYLOAD · RASPBERRY PI 4B · NEVER TRANSMITS

- Low-cost radio: cell tower sync signals
- GPS receiver with timing pulse
- Motion sensor
- Barometer

↓ Authenticated link, or a recorded session ↓

LAPTOP · ONE CODE PATH FOR SIMULATION, RECORDINGS AND LIVE

1 Hear towers
Arrival time of each cell's sync signal

2 Look them up
The public Canadian tower registry

3 Fuse
A position with a bound under our model

4 Tell the operator
Green, amber or red

Every number we show is labelled simulated or real

LIVE DEMO · SIMULATION

The same attack, without our jump guard and with it

1 · GPS LOST

Red within 5 seconds

The panel says: do not navigate on this position.

2 · SPOOFED, NO GUARD

Caught, then green and wrong

Half a minute later the plain filter takes the fake GPS back.

3 · SPOOFED, WITH GUARD

Stays red, then recovers

The fake GPS stays out. When the spoofer stops, real GPS comes straight back.

Simulated drone over the real tower sites around Carleton · one run · the guard is an option, not our default

REAL SIGNALS

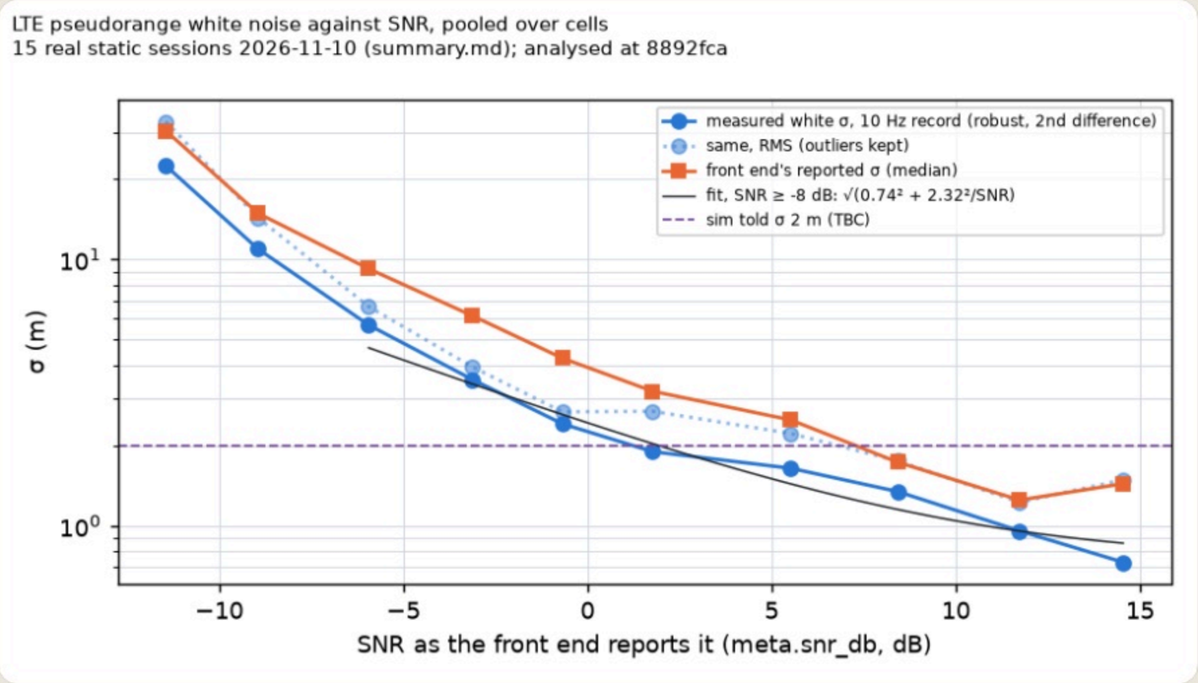
On real towers our radio is precise, a slower error hides underneath, and real position is not there yet

0.73 m

Range jitter on the strongest cells, measured on real towers.

But a slower error of about 8 m sits on top, and the radio does not report it.

Four real sessions, GPS removed in software, standing still: the towers made position hundreds of metres to kilometres worse, against about 2 m without them. Nothing is measured on the move yet.



WHERE WE MISS, AND WHAT CLOSES THE GAP

Detection works. Accuracy without GPS does not yet: the filter over-trusts the towers

Works

GPS jumps and a drifting GPS clock caught within seconds. No false alarms on clean runs.

Does not yet

Accuracy without GPS: no simulated flight meets 25 m, or the mentors' 50 m fallback.

What closes the gap

Learn each tower's real noise, and hop across 13 carriers. Our payload cannot hop yet.

The error that 95 % of 50 simulated flights are under at every second of the first 5 minutes without GPS (log scale)



Dashed line: our 25 m target. No row reaches it yet.

Simulation · against our own targets · the last row is a study: our payload cannot hop yet

OUR DESIGN RULE

Every source can lie, GPS and the towers alike

Source	How it can lie
GPS	A jump, a slow drag, fake time, a fake sky
Cell towers	A clock jump, a wrong registry entry, a fake tower
Aircraft reports	They carry someone else’s GPS

Towers take their time from GPS: by design they may contradict GPS time, but never confirm it.

Real recording, GPS sky faked in software: 59 of 59 fake sky reports flagged, no false alarm on 120 real ones.

The bound, on two real sessions: the default filter’s was wrong on 34 % of 3,005 checks; our opt-in honest-bound filter’s on none. It is kilometres wide: honest, not yet usable.

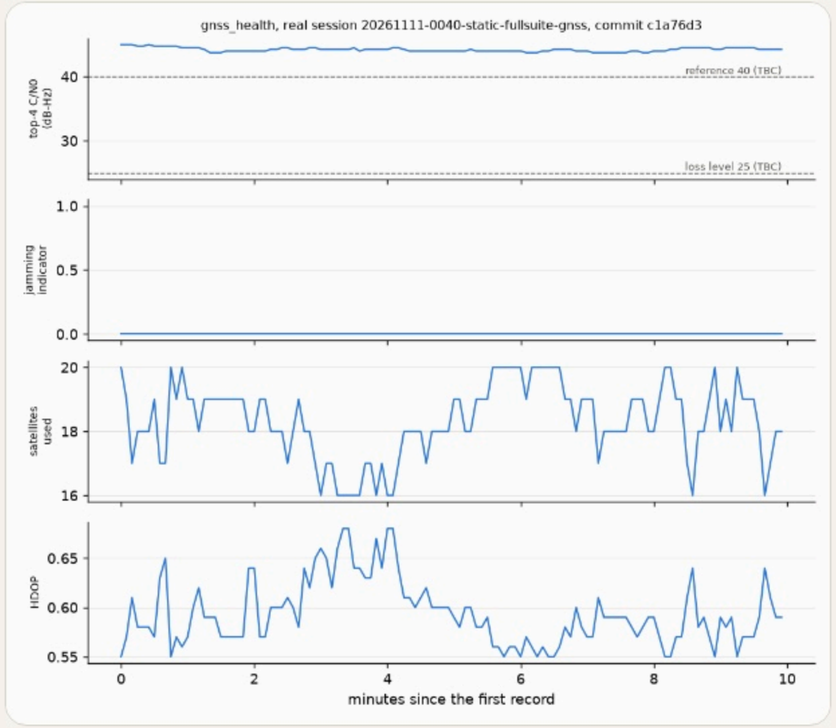


Figure: real GPS signal power from our payload, one of the cues the sky check reads

ROADMAP

The product: a receive-only GPS trust monitor that warns the autopilot

PRODUCT

It checks GPS against cell towers

Today it flags on a ground laptop. The output to the autopilot is not built.

THE GAP WE FOUND

Nobody sells this check

We found no spoof detector on sale that checks GPS against cell towers, and no vendor publishing false alarms per flight hour.

PROOF IT STILL NEEDS

Three measurements

1. Detection on recorded spoofing
2. False alarms per hour over clean flight hours
3. One authorised live-jamming event

First buyers, proposed, none approached: operators flying beyond line of sight, drone shows, a defence test slot.

Cost: radios and antennas under CAD 170. The full payload is not yet priced, weighed or power-measured.

Thank you